National Conference on Communications (NCC)

CEDAT, Makerere University

MajiQuick: A Solar-Powered Smart Water Access System for Sustainable Urban Communities

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Abstract

Households in Kampala's informal settlements buy water at vendor-operated taps where every transaction is settled in cash and the vendor must be physically present. Service hours are limited by that presence, and 30–40% of revenue is lost to unaccounted-for water.

MajiQuick decouples payment from collection. Credit is bought remotely over USSD or a web application, and a six-digit code delivered by SMS carries a balance in litres that is drawn down over several visits. The tap dispenses automatically, runs on solar power and needs no attendant, no smartcard and no internet connection.

Problem Statement

- Over 60% of residents of Kampala's informal areas have no direct piped connection [1].
- The tap functions only while the vendor is present, so service hours are set by availability rather than demand.
- No automated record links water dispensed to cash collected; NWSC estimates 30–40% leakage [2].
- Women and girls make roughly 70% of collection trips, each taking 30–60 minutes [3].

Objective: separate paying for water from being physically present at the point of collection, using only technologies these households already own.

System Architecture

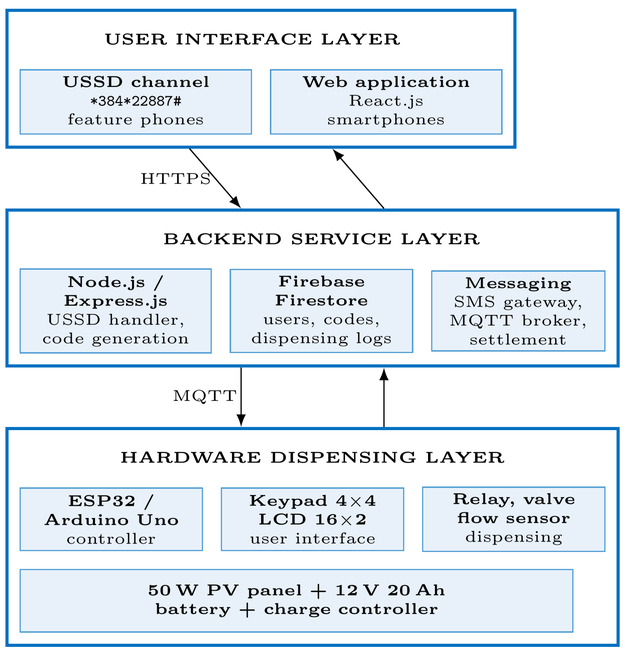


Fig. 1: Three-layer system architecture.

Three independently replaceable layers:

- User interface — a USSD menu (*384*22887#) for feature phones and a React.js web application for smartphones.
- Backend — Node.js and Express.js with Firebase Firestore. Access codes are hashed with bcrypt, so the plaintext code is never stored.
- Hardware — ESP32 or Arduino Uno, 4×4 keypad, 16×2 LCD, Hall-effect flow sensor at 450 pulses/litre [5], relay-driven solenoid valve, 50 W PV panel and 12 V 20 Ah battery.

The balance is held in litres, not as a count of jerrycans, so one purchase supports repeated draws of 5, 10 or 20 L. Only one SMS is needed per purchase instead of one per collection, and the code and balance are cached at the tap so dispensing continues when connectivity drops.

Hardware and Operation

Hardware behaviour was validated in two circuit simulators: Wokwi for the ESP32 and MQTT path, and Proteus for offline Arduino operation under solar supply. At the tap the user enters the code, selects a draw volume, and the flow sensor counts pulses until the target count or the safety timeout is reached. A 20 L draw is 9,000 pulses, 240 s at 5 L/min, with a 300 s safety cutoff. The photovoltaic supply is sized on 4.5 peak sun hours per day for central Uganda, taken from measured national irradiance data [6].

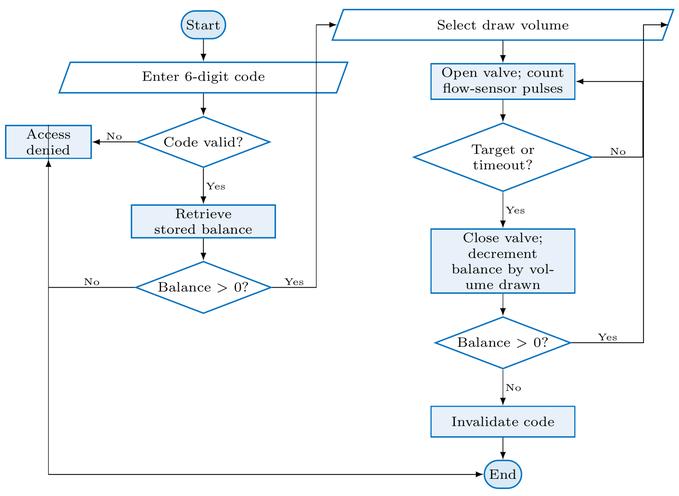


Fig. 2: Dispensing and balance decrement at the tap.

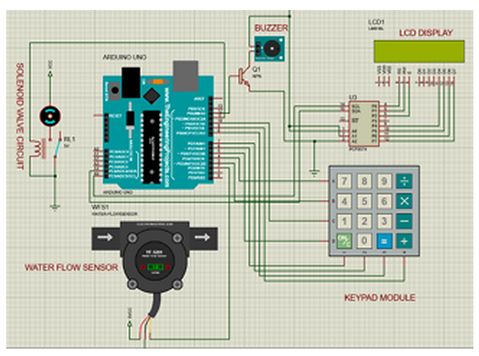


Fig. 3: Proteus circuit model of the dispensing unit.

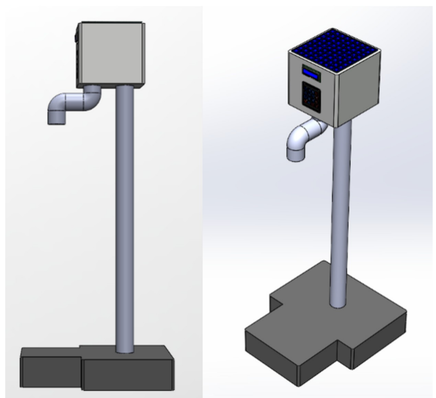


Fig. 4: SolidWorks enclosure model.

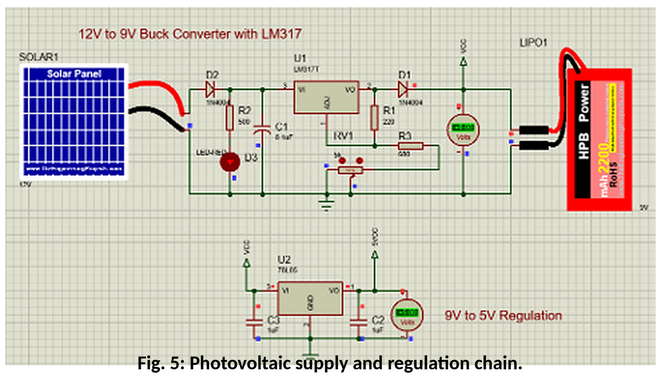


Fig. 5: Photovoltaic supply and regulation chain.

Results

All 15 functional test cases passed, covering the USSD and web purchase flows, code validation, dispensing behaviour and multi-draw balance handling, with valid, invalid and exhausted codes exercised in both the online (Wokwi/MQTT) and offline (Proteus) configurations.

USSD menu response averaged 0.8 s against a 3 s usability target, code generation 0.4 s, and SMS delivery 4.2 s. A single code sustained eight sequential draws before being invalidated automatically.

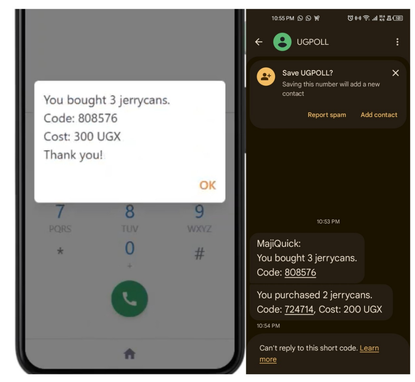


Fig. 6: USSD purchase and SMS access code.

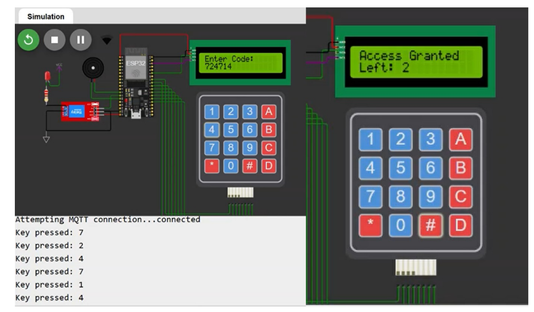


Fig. 7: Valid code accepted at the tap.

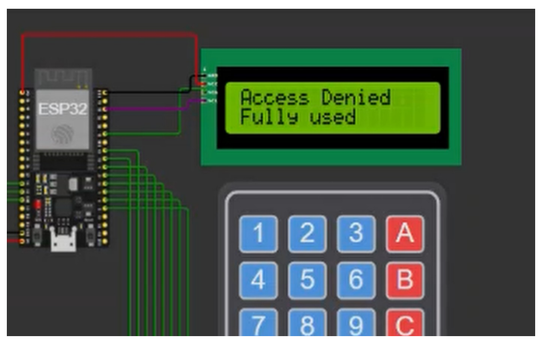


Fig. 8: Exhausted code refused.

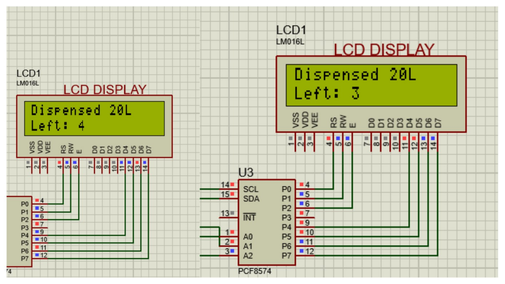


Fig. 9: Balance decremented after a draw.

Quantitative Outcomes

Metric	Result
Daily energy demand	40 Wh (nominal) / 68 Wh (high demand)
Battery autonomy at 50% DoD	3.0 days / 1.8 days
PV yield, 50 W at 4.5 PSH	157 Wh/day — 2.3× margin at high demand
Capital cost per tap	UGX 774,000 (≈ USD 206)
Payback period	3.6 months central, 8.1 months pessimistic
Messaging cost vs single-use codes	Reduced 8×, UGX 75,000 → 9,400 per month
Transaction overhead reduction	90.5–98.3% (analytical model)
Functional test cases passed	15 of 15

Conclusions

The reusable volumetric access code is the design choice with the widest consequences. It lowers messaging cost eightfold relative to single-use tokens, supports partial draws, matches how households already budget for water, tolerates loss of connectivity, and removes the physical smartcard or NFC token that every deployed prepaid system in the region requires [4].

The evidence is simulation-based and the conclusions are bounded accordingly. The results establish that the design is internally consistent and technically feasible: the energy budget closes with margin under Kampala irradiance, the dispensing logic behaves correctly across valid, invalid and exhausted codes, and payback is under one year in every scenario considered. They do not establish field performance — no hardware has been built, no real payment processed and no user observed.

Next: integrate the MTN MoMo and Airtel Money APIs, build a physical unit, add rate limiting on code validation, and pilot five taps in a Kampala settlement with at least 100 participants.

References

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